

Referee Report

Communication Methods and Measures

Prepared July 26, 2026

Manuscript title: *“Using Digital Twin Simulation for Synchronous Communication Research: With an Application to the Structured Group Design”*

Recommendation: **Major Revision**

Summary of the Submission

The manuscript proposes two linked contributions. First, it introduces the “structured group design,” a novel synchronous discussion protocol in which every participant responds simultaneously at each of three stages (initial standpoint and reasoning, counterargument, response to counterargument), with an algorithmic pipeline summarizing responses between stages. The design is presented as a scalable alternative to the traditional small-N, turn-taking focus group or deliberative session, which the authors argue is vulnerable to unrepresentativeness, path dependence, and groupthink. Second, and more central to this journal’s mission, the manuscript proposes a general method — LLM-based digital twin simulation — for evaluating the properties of any synchronous communication design before (or instead of) fielding it with human participants. The authors simulate the structured group design at three scales ($N = 6, 60, 600$) across several topics (nuclear power, gun control, expansion of the U.S. House of Representatives), measure the stability of stage-3 summaries across repeated trials using pairwise cosine similarity of text embeddings, and argue that small-N trials produce unstable, exclusionary, “garden of forking paths” outcomes while large-N trials converge to a stable distribution. The paper further proposes two assumptions required for such simulations to license inference to population-level discourse (coverage and unbiasedness), together with four validation procedures (backward continuity, frontward continuity, a reference-list coverage check, and a bias audit), and reports a partial replication against a human dataset from an Ohio State deliberative town hall.

The topic is squarely within scope for *Communication Methods and Measures*: it is a methodological contribution addressing design validity and measurement in synchronous group communication research, and it engages seriously with validation and epistemology rather than simply showcasing an application. The writing is clear, the literature review is thorough and current, and the AI-disclosure appendix is a model of transparency that other authors using LLM methods should be encouraged to emulate. I also want to acknowledge that the authors have already built in more validation machinery (four distinct tests, a human-data replication, an explicit statement of assumptions) than is typical in this fast-moving area, and the paper is a genuine, good-faith attempt to bring rigor to a domain that badly needs it.

That said, I have a set of major concerns, detailed below, that center on (a) a gap between what is simulated and what is claimed about the traditional design, (b) the statistical treatment of the similarity data, and (c) the scope and rigor of the validation tests themselves. I think the contribution is publishable after these are addressed, but the revisions are substantive rather than cosmetic.

Major Comments

1. The central comparative claim is not directly tested. The paper’s motivation rests on contrasting the traditional small-N synchronous design (vulnerable to path dependence, groupthink, and turn-order effects because participants speak sequentially) with the proposed structured group design. But the simulations reported in the paper only vary N within the structured design itself (6 vs. 60 vs. 600); the traditional, turn-taking design is never simulated. The paper even reports (Appendix D) that the authors built an “unstructured group design” simulator as a demonstration of the pipeline’s generality, but explicitly declines to report results from it. As it stands, the finding that small samples are noisier than large samples is a general property of sampling variability that would arise under any design — structured or not — and the paper does not isolate what, if anything, is specifically attributable to the turn-taking/path-dependence mechanism that motivates the whole comparison. I would ask the authors either to simulate the unstructured/turn-taking design directly (even a stylized version) so the two designs can be compared under matched N, or to retitle and reframe the contribution more modestly as being about sample size and instrument design within the structured group design, with the case against the traditional design remaining a literature-supported hypothesis rather than a demonstrated result.

Thank you for this comment. We have removed the language about turn order effects and instead focus on the sample size design feature.

2. The proposed mechanism (path dependence, groupthink from sequential speech) is not actually modeled. Related to (1): nothing in the simulation pipeline represents sequential turn-taking, social influence, or dominance dynamics. Each persona responds independently to a prompt containing an aggregated, already-summarized “opposing viewpoint” — there is no order effect, no accumulating conversational history among individual (unsummarized) participants, and no mechanism by which one persona’s response could suppress or amplify another’s within a round. If the point is that small samples are unstable, that is a sampling argument; if the point is that turn-taking causes idiosyncratic path dependence, that requires a design that actually encodes sequence and influence. I recommend the authors be explicit about which claim the simulation supports and avoid language (e.g., “path dependence,” “groupthink”) that implies the sequential-influence mechanism has been simulated when only the sampling mechanism has.

Please see our previous response.

3. Non-independence of the pairwise cosine similarity data. The reported distributions (e.g., 435 pairwise comparisons among 30 trials at $N = 60$) are built from all pairwise combinations of a much smaller number of trials, so each trial contributes to many comparisons and the observations plotted in the figures are far from independent. Treating these as if they were 435 (or 1,125, or 190) independent draws when characterizing “modes” and dispersion risks overstating precision and understates uncertainty. I would ask for a trial-level (rather than pair-level) analysis — for instance, a permutation test, a mixed-effects model with trial random effects, or bootstrap resampling of whole trials — to support any claims about the shape or stability of these distributions, particularly since the paper leans on visual inspection of density plots (“mode at zero,” “unimodal at 0.9”) rather than formal tests.

This is a very helpful comment. We have made two changes to accommodate this issue of dependence in the cosine similarity scores. First, we have changed the main figures to histograms, where we use a color scheme to emphasize that the trials that produce empty texts, which causes a deterministic zero for cosine

similarity with all other trials. In the new figures, it is clear the histogram represents a mixture distribution, where the cosine similarity is either zero or non-zero, and if it is non-zero the histogram shows the conditional marginal distribution. The marginal distribution is not affected by dependence provided there are either many trials, or if dependence does not create clustering that would distort the distribution. To show this, in the appendix we provide MDS plots showing that the large-N designs do not have any clustering. The small-N design does have some clustering but we run the process to more than 1k comparisons.

4. The validation tests rest on very small, single-coder samples. Backward continuity is assessed from one research assistant's judgments on 60 observations from a single trial; frontward continuity from one research assistant reading one 60-participant trial; the Britannica coverage check from one human coder and one LLM coder disagreeing on 22 of 60 items, adjudicated post hoc by the lab team rather than by a blinded third party. There is no inter-rater reliability statistic (e.g., Krippendorff's alpha) reported anywhere, and four separate significance tests are reported for backward continuity without correction for multiple comparisons. Given that a communication-methods audience will scrutinize measurement rigor closely, I would ask for (a) at least two independent coders per validation task with a reported reliability statistic, (b) validation across more than one trial per topic (the paper asserts that "since each trial is a random draw... it does not matter which trial we investigate," which is precisely the claim the paper's own instability results call into question for small samples), and (c) a multiple-comparisons correction or a single omnibus test where appropriate.

The backward and frontward continuity tests are not essential for the paper. These tests are not related to the two core assumptions (coverage and unbiasedness). The frontward and backward continuity tests are just intended as a part of the human oversight of the LLM output. We have moved those to the appendix to help clarify that. We also have scaled the coverage test coding to a total of six sessions, and human labeling of 180 labels is also part of the human oversight of that process.

5. The human-data replication answers a different validity question than the one advertised. The Ohio State IDEA town hall data are post-discussion, individual open-ended survey responses collected after a synchronous deliberative event — not data generated by the three-stage structured group design itself. Showing that the LLM persona/summarization pipeline can approximate the argument space found in independently collected human open-ended text is a reasonable and useful check on the summarization and persona-generation machinery, but it is not a validation of the structured group design's three-stage exposure-and-counterargument mechanism, which is the paper's other central contribution. I would ask the authors to state explicitly what this replication does and does not validate, and, if possible, to identify or collect a human dataset that was itself generated under something closer to the three-stage protocol.

We are clear that the OSU town hall was largely similar to the structured group design but we also are transparent about how it is different.

6. Embedding model and summary-length choices are unexamined sources of researcher degrees of freedom. All similarity results depend on a single sentence-embedding model (all-MiniLM-L6-v2) and on summaries fixed at five perspectives per option; the bias-audit metric (maximum cosine similarity to one of five summary sentences) will mechanically shift if the number of summary sentences changes. Because these choices are load-bearing for the paper's central quantitative claims (e.g., similarity of ~0.7 vs. ~0.9), I would like to see a robustness check with at least one alternative

embedding model and one alternative summary length, and some independent (human-judgment) anchoring of what a given cosine similarity value means substantively — e.g., do human raters agree that pairs scored around 0.7 represent meaningfully different arguments while pairs around 0.9 do not?

We typically see papers where the author uses a single embedding model. And the five perspective summary is constant across our trials. We think these choices are defensible.

7. Persona covariate sets appear to have been tuned to produce diversity, which should be discussed as a limitation rather than a footnote. Footnote 23 discloses that need-for-cognition and vertical-collectivism covariates were added to the House-expansion personas specifically because, without them, “liberal LLMs almost always chose to expand and conservative LLMs almost always chose to keep the House the same.” This is a commendably transparent disclosure, but it also means the coverage results are sensitive to which covariates the researcher happens to add, and the paper does not offer general guidance for how another researcher should choose or validate a covariate set for a new topic (beyond “keep adding covariates until the LLM personas diversify”). Given that the coverage assumption is one of only two assumptions the paper says are strictly required, this deserves more sustained discussion — ideally with a suggested procedure (e.g., pre-registering covariates against a known distribution of positions, or a sensitivity analysis varying the covariate set) rather than an ad hoc, outcome-driven addition of covariates.

We have added text to that footnote calling attention to the script in the repository that is designed to allow researchers to experiment with covariates to create an optimal set or to try to warrant meeting the distribution match assumption.

8. Pooling across four LLM back-ends may conflate model-specific artifacts with population diversity. Each persona is assigned one of four LLMs (Gemini, Qwen, Llama, GPT-OSS) held constant across its three rounds, but the paper does not decompose how much of the between-persona variance in the pooled results is attributable to genuine covariate-driven diversity versus systematic differences in the four models’ response tendencies (a concern the paper itself cites extensively in the bias literature). The appendix bias audit hints at this — noting an over-representation of OSS-GPT responses — but this issue is not addressed for the main stability results in Section 6. A breakdown of the cosine-similarity results by LLM, or at least a single-LLM robustness run, would strengthen the claim that the observed diversity reflects the target population rather than an artifact of ensembling four heterogeneous models.

All of the results are marginal to the models, so we don’t think this is a significant issue. This would matter more if we were claiming to meet the faithfulness assumption, which would not be met if the model was responsible for variation in arguments. We also do a full replication of the results with different models to show the results are not model dependent.

9. The normative scope of the claim needs fuller treatment for a communication-methods readership. Footnote 3 concedes the structured group design is inappropriate when the research question concerns face-to-face interaction, emotion, or conflict, but this important scope condition is dispatched in a single sentence. Much of the communication field’s interest in synchronous group methods is precisely in the interactional, emic process (turn-taking, uptake, relational dynamics), not only in

aggregating stable summaries of positions. I would ask the authors to more fully situate the contribution relative to that literature and be clearer, throughout (not only in a footnote), about which research questions the method is, and is not, suited to answer.

We believe that we are clear about this limitation in the conclusion.

Minor Comments

1. The phrase “garden of forking paths” is strongly associated in the methods literature with Gelman and Loken’s critique of researcher degrees of freedom in statistical analysis. Its use here to describe path-dependent group dynamics is a different concept and may confuse methodologically minded readers; consider a different label or an explicit disambiguating clause at first use.

We eliminated this term when we removed the discussion about the pathologies of turn taking in discussion (see points 1 and 2 in the major comments above).

2. The abstract emphasizes the empirical finding (small-N designs are unstable) more than the more portable methodological contribution (the coverage/unbiasedness assumption framework and the four-part validation toolkit), which is likely to be of greater interest to this journal’s readership. Consider rebalancing the abstract accordingly.

We have revised the abstract in this way.

3. A summary table listing the four candidate assumptions (coverage, unbiasedness, faithfulness, distribution match), which are required, and which validation test addresses which assumption, would help readers track an otherwise dense conceptual section (Section 8).

We now only discuss the coverage and unbiasedness assumptions with reference to the validation tests, and since there are only two relevant assumptions we don’t think a table is necessary.

4. The paper never offers an interpretation for why the “prioritize researching improved implementations” option was so dominant among LLM personas on the nuclear power topic. Given that this skew required the authors to drop the option in a follow-up run, a sentence considering whether this reflects a known LLM tendency toward hedged, non-committal responses would help readers anticipate similar artifacts in their own applications.

We have moved these results to the appendix, and we note in the main text is that it is the tendency of LLMs to hedge in their response that causes the skewed response data.

5. The convention of setting cosine similarity to zero whenever a summary is empty (footnote 14) deserves more justification, particularly for the empty-versus-empty case: two trials that both consistently exclude an unpopular option are arguably showing agreement (a consistent exclusion), not instability, yet they are coded identically to a trial with an empty summary compared against a trial with a substantive one. This coding choice may inflate the “mode at zero” in ways worth disentangling in a footnote or robustness check.

We think it is defensible that whenever a text cell is empty, its similarity to other trials is zero. We have clarified the figures that the mode at zero is driven by the deterministic coding of these trials to zero.

6. Section 9's classification of which human arguments were or were not "replicated" by the LLM personas (6 of 7 in favor, 4 of 7 opposed) appears to be an author-team judgment call rather than an independently coded, reliability-checked classification; given how central this comparison is to the paper's validation case, an independent or blinded coding pass would strengthen it.

This table is fully replicated in the appendix, and the text is short enough that a reader can verify our coding for themselves.

7. Please report summary statistics (means, SDs, Ns per condition) for the cosine-similarity distributions in the main text or a table, in addition to the density figures, so that readers and future meta-analysts are not limited to visual inspection of plots.

We have added these summary statistics to the figures and we report them as well in tables in the appendix for both the original experiments and the replications.

8. Footnote 15's claim that "190 trials is a sufficient number" for the large-N condition is asserted rather than justified; a brief power/precision rationale (e.g., a stability check across an increasing number of trials) would substantiate this.

We have added text in this footnote to better justify this.

9. Given that this is fundamentally an infrastructure and methods paper (Prytaneum), the discussion would benefit from a brief, concrete treatment of the practical costs and moderation demands of running a real N = 600 structured group session with human participants, so applied readers can weigh this design against, e.g., a large-N survey with open-ended follow-up or existing AI-mediated deliberation platforms (e.g., Tessler et al., 2024; Lu et al., 2024, already cited) on cost and fidelity grounds.

Footnote 2 has a discussion of human data collection costs.

10. Minor: the manuscript could more explicitly differentiate the "structured group design" from other recent AI-mediated deliberation or discussion platforms already cited (Tessler et al., 2024; Lu et al., 2024; Zhang et al., 2024) with a short comparison table, to sharpen the design contribution for readers less immersed in the deliberation literature.

This seems outside of the scope of this paper.

Recommendation

This is a serious, transparent, and well-motivated contribution to an important and under-addressed problem — how to validate design choices for synchronous group communication research using LLM simulation — and it is a good fit for this journal. I recommend major revision. The authors should (1) either simulate the traditional/unstructured design directly for a matched comparison or reframe the claim as being about sample size rather than design type, (2) address the non-independence of the pairwise

similarity data with a trial-level analytic approach, (3) strengthen the validation tests with multiple coders, reliability statistics, and more than a single validated trial per topic, and (4) add robustness checks around the embedding model, summary length, and LLM ensemble choices that currently function as unexamined researcher degrees of freedom. I would be glad to review a revised version.